Pitch
See BASIC CONTROLS *panel, p.429.*

Pitot tube
A small, open-ended tube, usually located on the leading edge of the wing, which is used to measure airspeed and is linked to a pressure-measuring device in the cockpit.

Powerplant
The permanent assembly (including propellers and engine) which is responsible for aircraft propulsion.

Pressurized
Describes a cockpit or cabin that maintains a constant pressure, regardless of the aircraft's altitude.

Propeller
A device consisting of airfoil-shaped blades that spin around a central hub. As the engine turns the blades, they create thrust by "biting" into the air and forcing it to move back.

Pusher
A configuration in which the propeller is situated at the back of the engine or fuselage and pushes the aircraft through the air.

Radar
Radar (radio direction and ranging) is an instrument used to locate an object, to navigate, and in warning and detection systems. Radar works by transmitting radio waves and noting the time it takes for the reflected waves to return.

Radar signature
The waveform of a radar echo, which can be used to identify an object.

Ramp doors
The doors inside the intake ramps of a jet aircraft's engine. These regulate the flow of air to the engine.

Radial engine
See PISTON ENGINES *panel, left.*

Radome
A protective radar dome – which allows radio waves in and out – covering radar equipment.

Roll
See BASIC CONTROLS *panel, p.429.*

Rotary engine
See PISTON ENGINES *panel, left.*

Rotocraft
See HELICOPTER *panel, right.*

Rotors
Airfoils, used on a helicopter, that rotate at high speed to create thrust and lift.

Ruddervator
A movable structure that combines the functions of elevator and rudder.

Rudder
A control surface on the fin's trailing edge which governs yaw.

SAM (surface-to-air missile)
A missile fired at an enemy aircraft or missile, launched from the ground or from a ship.

Satellite
A spacecraft or other body orbiting the Earth.

Seaplane
An aircraft that can land on or take off from water.

Skid
Sled runners used as part of an aircraft's undercarriage.

Smart weapon
A precision-guided munition that is directed to its target using laser guidance or, more recently, satellite-linked GPS (global positioning system) guidance.

Sonic boom
The thunderlike "clap" heard when an aircraft flies faster than the speed of sound. This is caused by sudden changes in air pressure as the craft pushes air molecules out of its path.

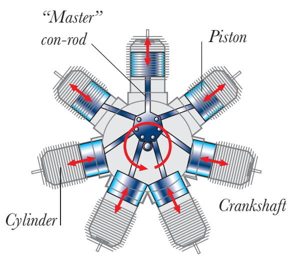
Spar
The main structural element that run the span of a wing. The ribs and other secondary structural parts are attached to the spars.

Splitter plate
A plate located between a jet engine's air intake and the fuselage, which evens out the air flow to the engine.

SST (Supersonic transport)
Commercial airliners that can carry passengers at supersonic speeds.

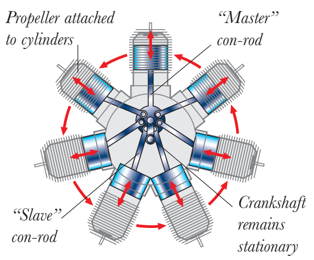
Stall
See AIRFOIL *panel, p.429.*

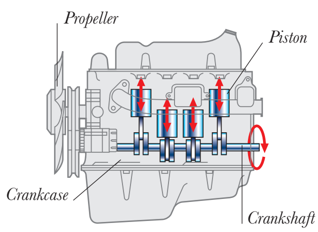
PISTON ENGINES



RADIAL ENGINE
A radial engine has an odd number of cylinders arranged in a circle around the crankshaft. The "master" con-rod is attached to the crankshaft while the other con-rods are "slaved" off the master rod. The propeller is attached to the rotating crankshaft.

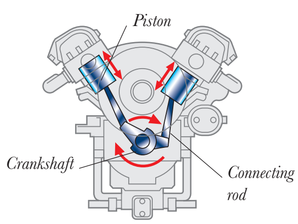
ROTARY ENGINE
Similar to the radial engine, except that the crankshaft remains stationary while the cylinders and propeller rotate around it, cooling themselves in the process.

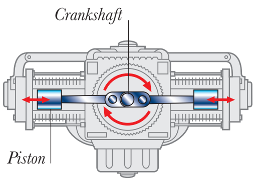




INLINE ENGINE
An inline engine has the cylinders in a line – with pistons pumping up and down inside – one behind the other in the crankcase.

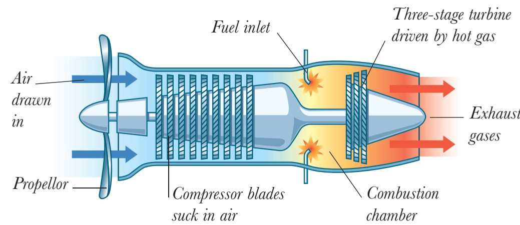
V ENGINE
Two inline banks of cylinders are arranged at an angle to each other. Each pair of pistons pumps diagonally on the same crankshaft, spinning it round.



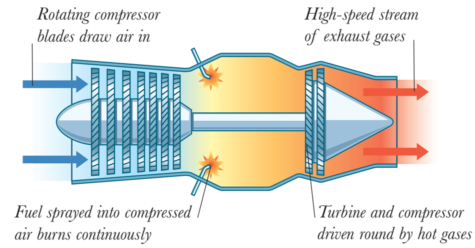


HORIZONTALLY OPPOSED ENGINE
Also known as the boxer engine, it has two banks of inline cylinders arranged opposite each other. This type is often used on trainers and small aircraft.

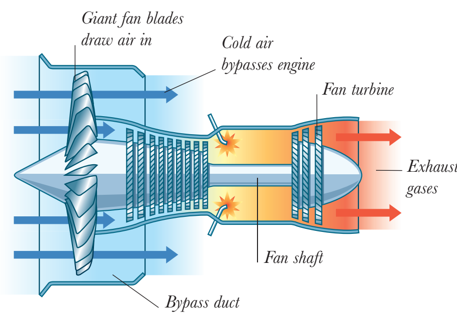
JET ENGINES



TURBOPROP
Fast-spinning compressor blades draw air into the combustion chamber, where it is heated by burning fuel. As expanding gases escape through the exhaust, they spin a turbine which powers the propeller.



TURBOJET
The simplest jets – "turbojets" – work by pushing a stream of hot exhaust gases out the back. This hits the air behind so fast that the reaction thrusts the plane forward like a deflating balloon.



TURBOFAN
The turbofan is quieter and more fuel-efficient than the pure jet or turboprop at airliner speeds (high subsonic) because of the combination of cold and hot thrust. The bypass air helps to keep the engine cool.